

Process Book

Justice Through Data: Visualizing Sentencing and
Recidivism Trends in California

Presented by

Shachi Benara, Julia Epshtein, and Jayani Tripathi

Jayani Tripathi

- **Email Addresses:** jayanitripat@umass.edu
- **SPIRE IDs:** 33292946

Shachi Benara

- **Email Addresses:** sbenara@umass.edu
- **SPIRE IDs:** 33353810

Julia Epshtein

- **Email Addresses:** jepshtein@umass.edu
- **SPIRE IDs:** 33180079

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INTRODUCTION

1.1 Overview and Motivation

Project Title:

Justice Through Data: Visualizing Sentencing and Recidivism Trends in California

Background and Motivation:

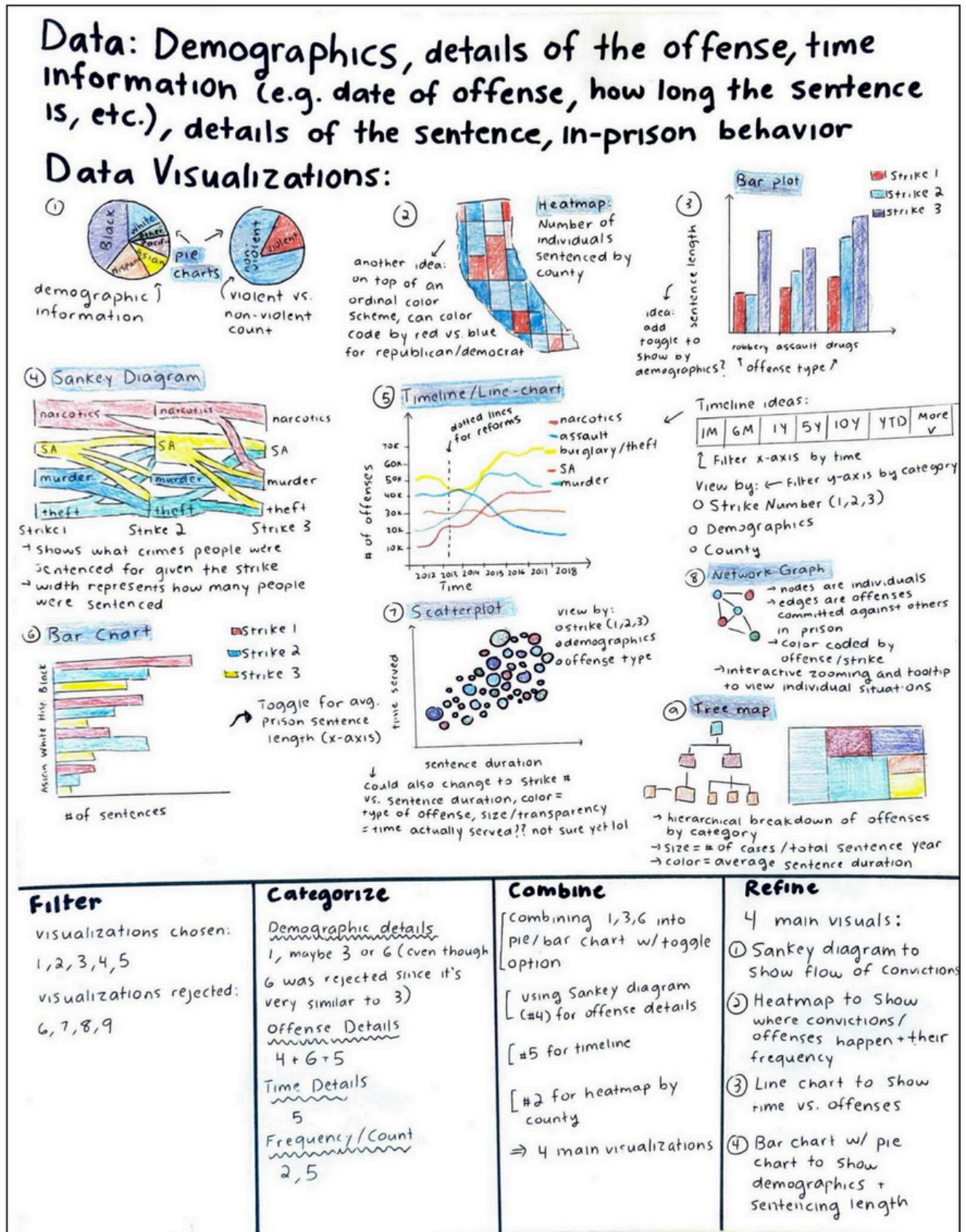
California's "Three Strikes and You're Out" law was designed to deter repeat offenders by mandating severe sentences—typically a minimum of 25 years to life in prison—for individuals convicted of three serious felonies. Initially intended to target violent criminals, the law has resulted in life sentences being imposed on many who committed nonviolent offenses. The real-world application of the law has led to unintended consequences, such as over-incarceration and a dramatic increase in the state's prison population.

The motivation behind this project is to critically examine these sentencing patterns and reveal how they diverge from the law's original intent. In doing so, we aim to expose how the law disproportionately impacts marginalized communities, including African American populations as well as individuals with mental or physical disabilities. By visualizing this complex data, we hope to contribute to the conversation on justice reform and highlight areas where current policies may require significant reconsideration.



INTRODUCTION

Our Initial Sketch



INTRODUCTION

1.2 Project Goals and Objective

The goal of this project is two-fold:

1. Awareness:

- a. **Objective:** Visualize the sentencing patterns under the Three Strikes law.
- b. **Approach:** Create a clear and interactive visualizations that not only present the data but also reveal anomalies—such as overly harsh sentences for nonviolent offenses—that highlight the law’s unintended consequences.
- c. **Outcome:** Enable the public to see how it impact on marginalized communities.

2. Advocacy:

- a. **Objective:** Use the data visualization to advocate for further reform and policy changes by visualizing how the current reforms have impacted the resulting prison population.
- b. **Approach:** Showcase how initiatives like Proposition 36, Proposition 47, and Prosecutor-Initiated Resentencing have altered sentencing trends, comparing historical data with more recent changes.
- c. **Outcome:** Empower policymakers, legal advocates, and the community with data-driven evidence that can lead to further meaningful changes in criminal justice policy.

INTRODUCTION

1.3 What We Aim to Accomplish

What we aim to learn and accomplish:

The goal for us is to learn about the impact of the Three Strikes Law on California's prison system, by looking into how sentencing patterns vary by offense type and demographics. We want to gain a better understanding of the composition of people who are sentenced, and identify patterns that could indicate cases eligible for resentencing. Essentially, our goal is to present legal and incarceration data in the most clear and interactive format possible that will be accessible to other users and potentially even policymakers.

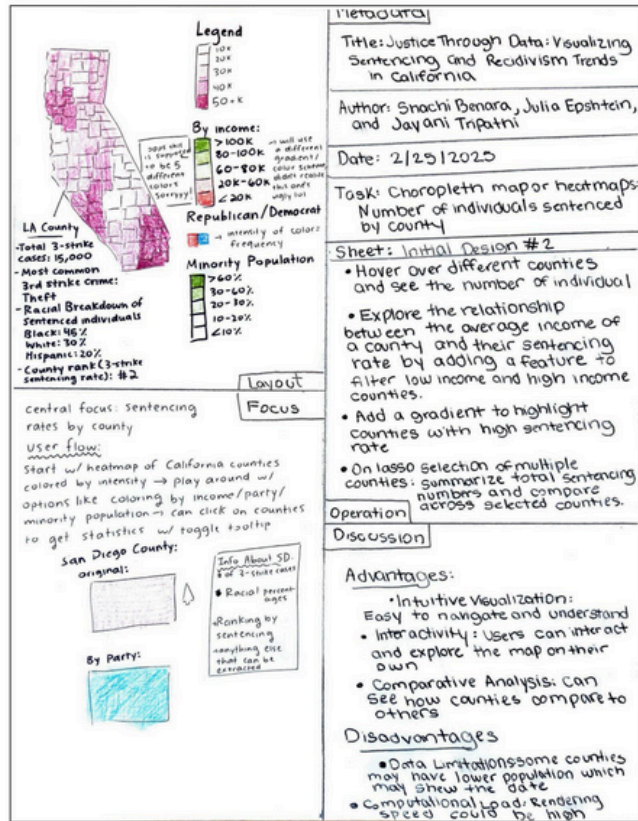
Main benefits of our goal:

We hope that our project will play an essential role in sentencing transparency, and supporting fairer judicial decisions using the information we visualized. The goal of this project is to identify unjust sentences, particularly cases where the individual receives harsher sentences for low-level offenses. The project can assist prosecutors, policymakers, researchers, and advocates in the justice system by providing visualizations. Our visualizations will highlight the harsh sentences given under this law, show biases in incarceration, and make the large and complicated data more accessible for the general public.

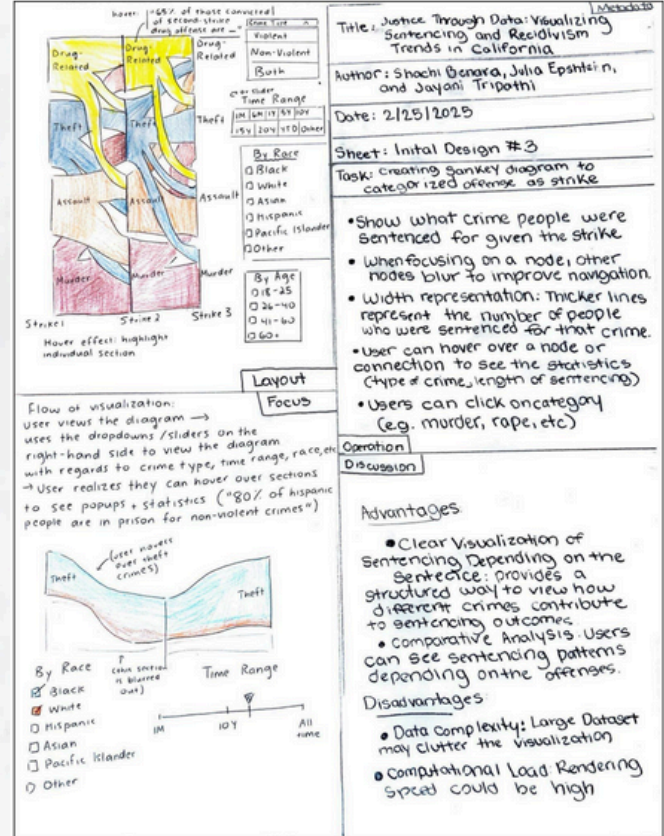
INTRODUCTION

Other Sketches:

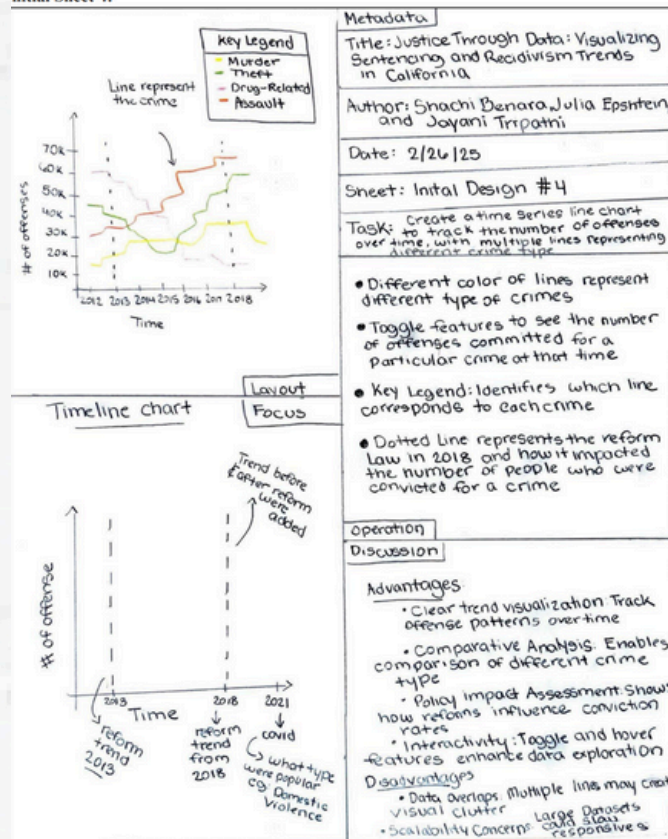
Initial Sheet 2:



Initial Sheet 3:



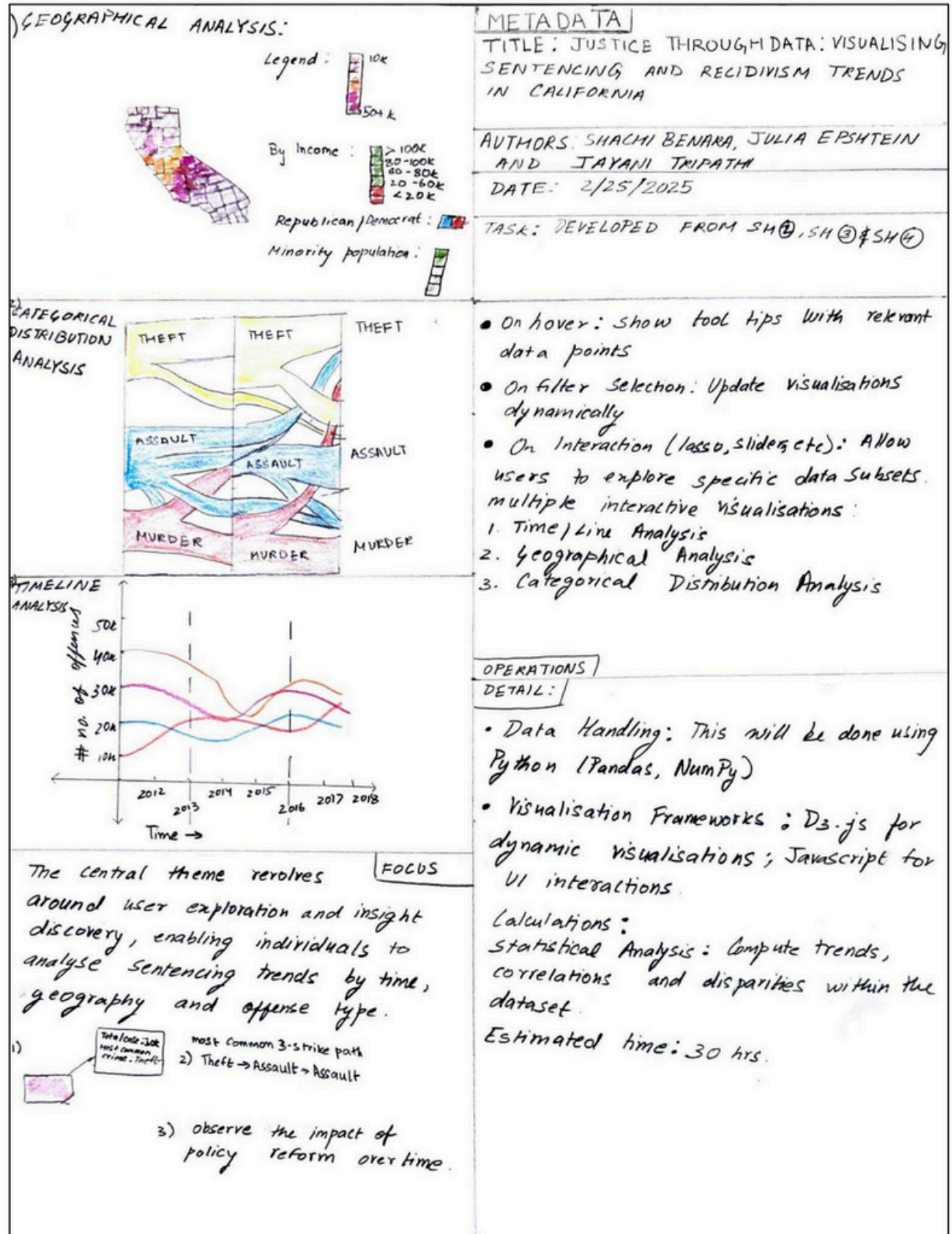
Initial Sheet 4:



INTRODUCTION

Final Sketch:

Final Sheet 5:



RELATED WORK

2.1 Inspirations and Background

Competition from MIT:

Our work was inspired by the MIT Policy Hackathon, where teams explored real-world policy issues using data. This event showed us how the data can be in driving change and how important it is to promote justice reform. In addition, we did a lot of online research, which gave us many ideas on how to present our data and tell our story clearly.

Research:

We took the time in researching online resources to find out how others have employed data to narrate stories about reform and justice. Learning this helped us figure out how best to break down complicated legal details into simple, interesting ways. We also did research on the California's law system.

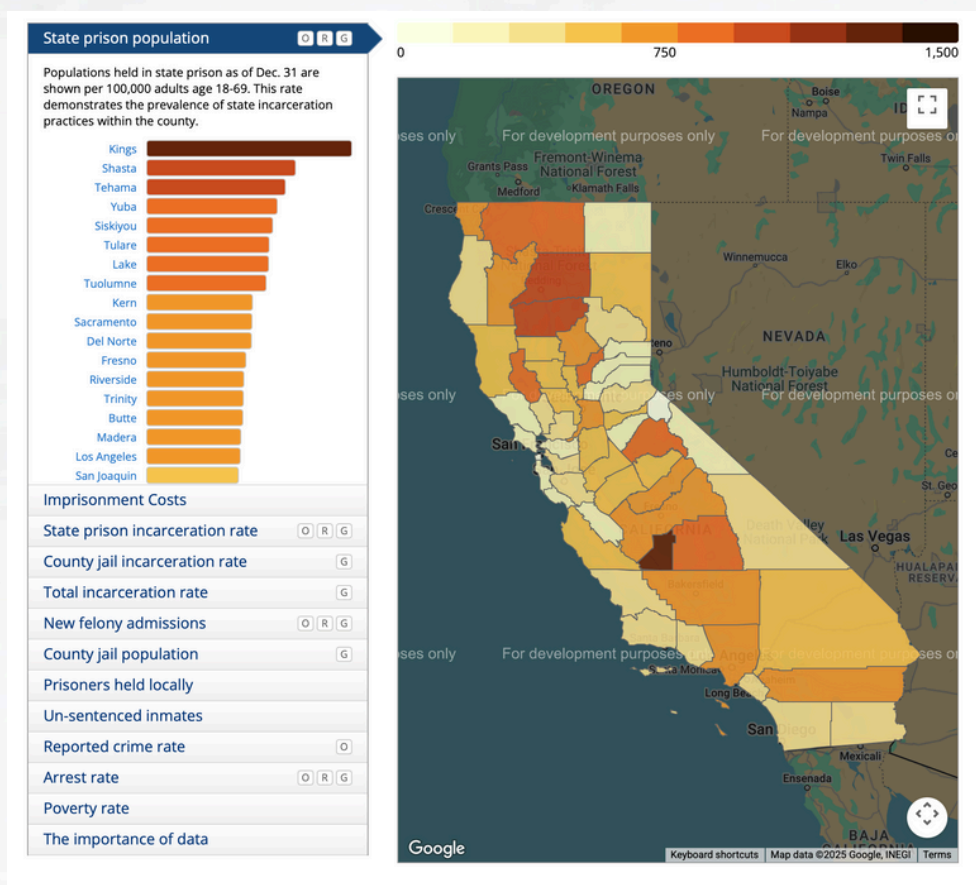
Reference:

- [1] Stanford Law School. law.stanford.edu/stanford-justice-advocacy-project/three-strikes-basics/.
- [2] California Courts. www.courts.ca.gov/documents/Prop47FAQs.pdf.
- [3] Cullen, F. T., Jonson, C. L., & Nagin, D. S. (2011). "Prisons Do Not Reduce Recidivism: The High Cost of Ignoring Science."
- [4] Hawken, Angela. "State of the Art of HOPE Probation." Hawaii State Judiciary, www.courts.state.hi.us/docs/news_and_reports_docs/State_of_the_Art_of_HOPE_Probation.pdf.
- [5] California Department of Corrections and Rehabilitation. www.cdcr.ca.gov/.
- [6] Redo.io. resentencing-data-initiative (Version 1.0.0) [Computer software]. github.com/redoio/resentencing_data_initiative/.
- [7] OECD (2019). The Path to Becoming a Data-Driven Public Sector. www.oecd.org/en/publications/the-path-to-becoming-a-data-driven-public-sector.

RELATED WORK

2.2 Existing Articles and Visualizations

We specifically looked at visualizations for California Sentencing rates and statistics and visualizations specifically for the California Three Strikes Law.



Link: <http://casi.cjcj.org>

RELATED WORK

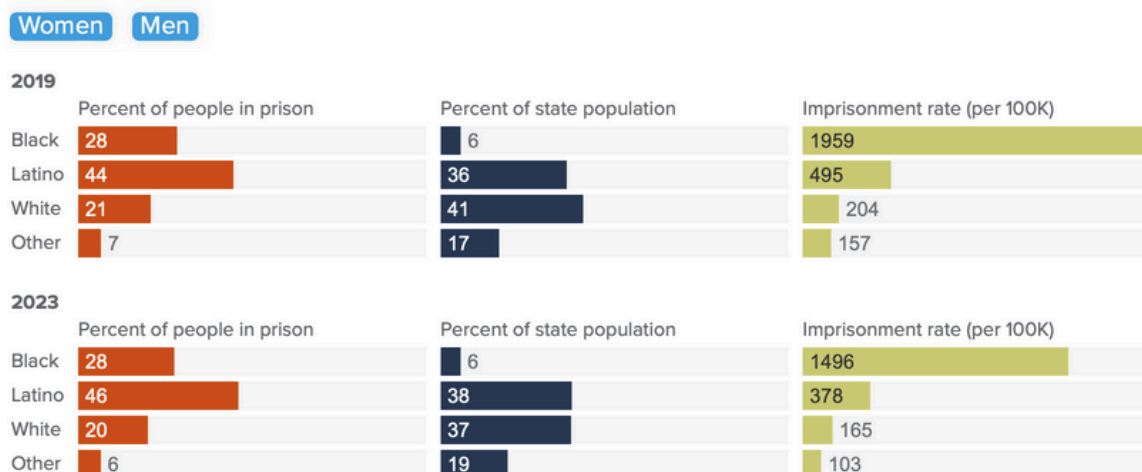
2.2 Existing Articles and Visualizations

We specifically looked for visualizations related to California sentencing rates and statistics, particularly those focused on the Three Strikes Law. However, we were only able to find visualizations of general prison sentencing rates in California, not ones specifically addressing the Three Strikes Law and its impact.

Link: <https://www.pplic.org/publication/californias-prison-population/>

California imprisonment rates fell during the pandemic and stark racial disparities persist

Click on the buttons to see data by gender:



SOURCE: CDCR Offender Data Points (December 2019 report, December 2023 dashboard) and CA DOF population data (2019, 2023).

NOTES: CDCR does not publicly provide disaggregated data for Asian Americans, Native Americans, and people of other racial groups. These groups are all included under "Other." Imprisonment rates reflect the number of people incarcerated in prison per 100,000 adult state residents.

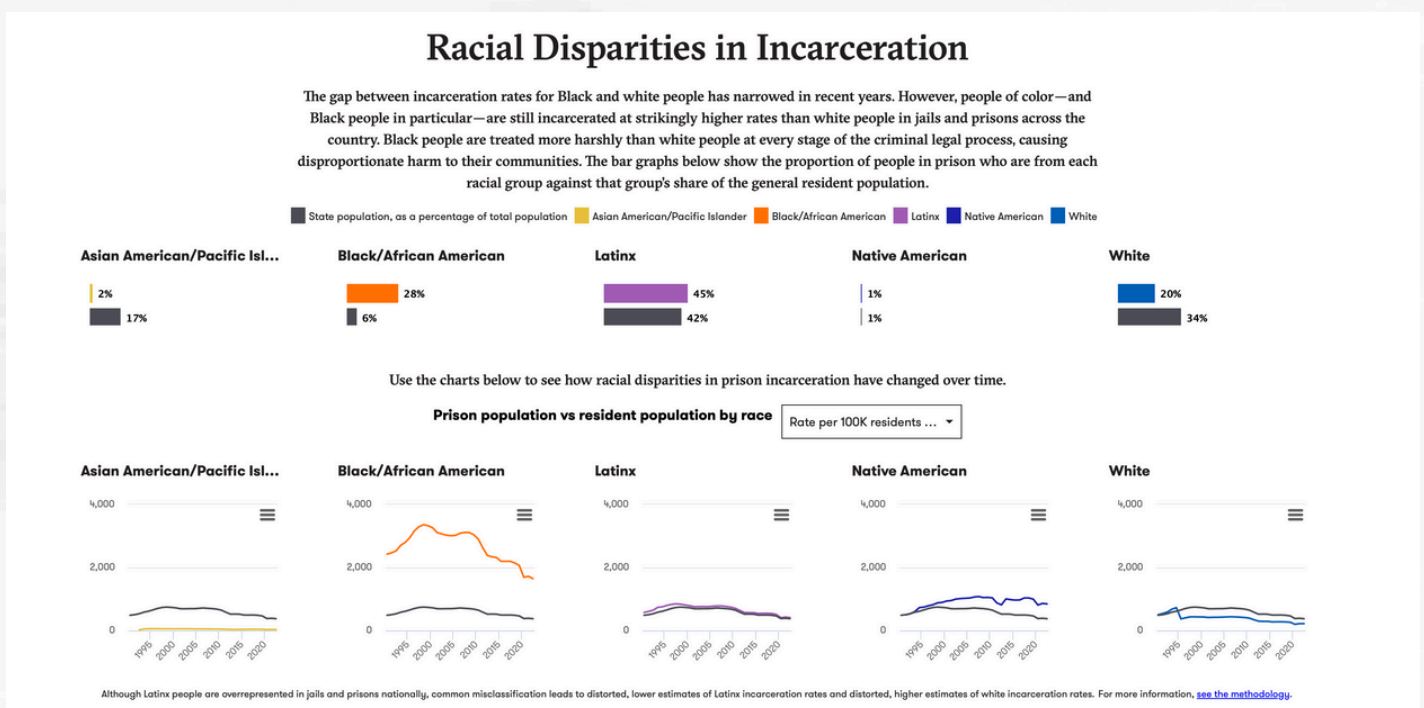
Chart: PPIC 2024 • Get the data • Embed • Download image

RELATED WORK

2.2 Existing Articles and Visualizations

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QUESTIONS

3.1 Research Questions

Primary Question: How can we use our visualizations to determine the impact and effectiveness of the Three Strikes Law

Other Question:

- Who is incarcerated in California prisons, and for what offenses?
- Does the dataset show any bias against certain demographics that should be taken in consideration when evaluating eligibility for justice reform measures?
- Are there certain regions that have higher sentencing rates?

3.2 Changing of Research Questions

We were initially concerned with broad, overall sentencing distributions. As we continued our inquiry, however, we encountered some serious data anomalies, such as:

- **Irregular Duration Values of Sentences:** Seeing numbers such as "1,000,000 months" compelled us to re-examine the numeric interpretation and hence handle certain variables as categorical.
- **Lack of "First Strike" Data:** Because the dataset proved to be incomplete, we revised our questions in order to better compare the existing data.

4. Tools

To decide which features from the data to use for our visualizations, we started by exploring the dataset in Python and Jupyter Notebooks. There are three main reasons we went with these tools for data cleaning and analysis.

1. These tools contain several built in libraries that allow us to quickly visualize the data (matplotlib and seaborn).
2. NumPy and pandas are powerful libraries that make it easy to manipulate, filter, and summarize large datasets efficiently.
3. Jupyter Notebooks makes it a lot easier to do exploratory data analysis as a group as it provides an interactive environment where we could combine code, easy-to-understand outputs, and written explanations in Markdown, which makes it easier to document our process and share our code.

DATA EXPLORATION

5. Initial Data Exploration

5.1 First Glance

- Used `.head()` to preview the first five rows in each dataset.

5.2 Missing Values

- Used `.isna().sum()` to tally missing values by column.

5.3 Unique Values

- For demographics: printed unique value counts for columns with fewer than 1000 unique entries.
- For commitments: examine uniqueness in key offense-related columns for both current and prior records.

5.4 Data Types

- Print `.dtypes` for each dataset to understand column types (e.g., object, datetime, float).

Observation

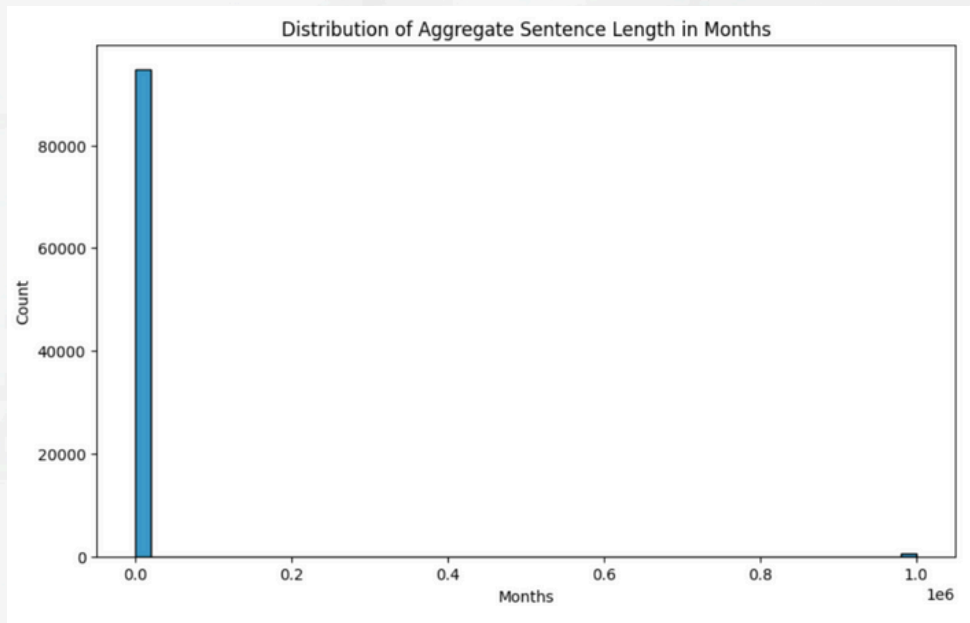
- `CDCNo` (inmate ID) is not unique in `current_commitments` → indicated multiple offenses per inmate.

DATA VISUALIZATION

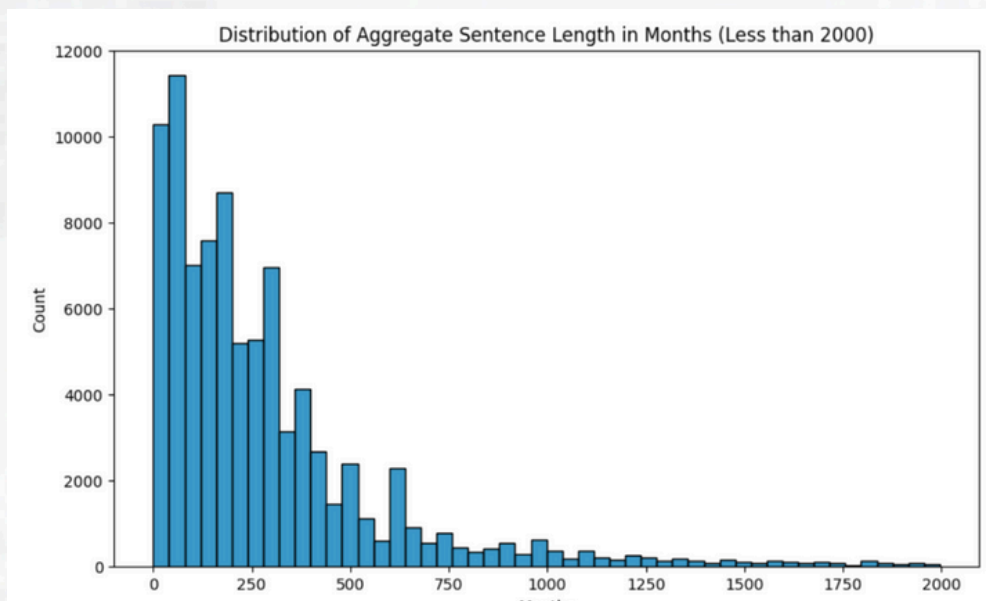
6. Data Visualization

6.1.1 Demographics

- Plot histogram of **Aggregate Sentence in Months**. Got this strange result:



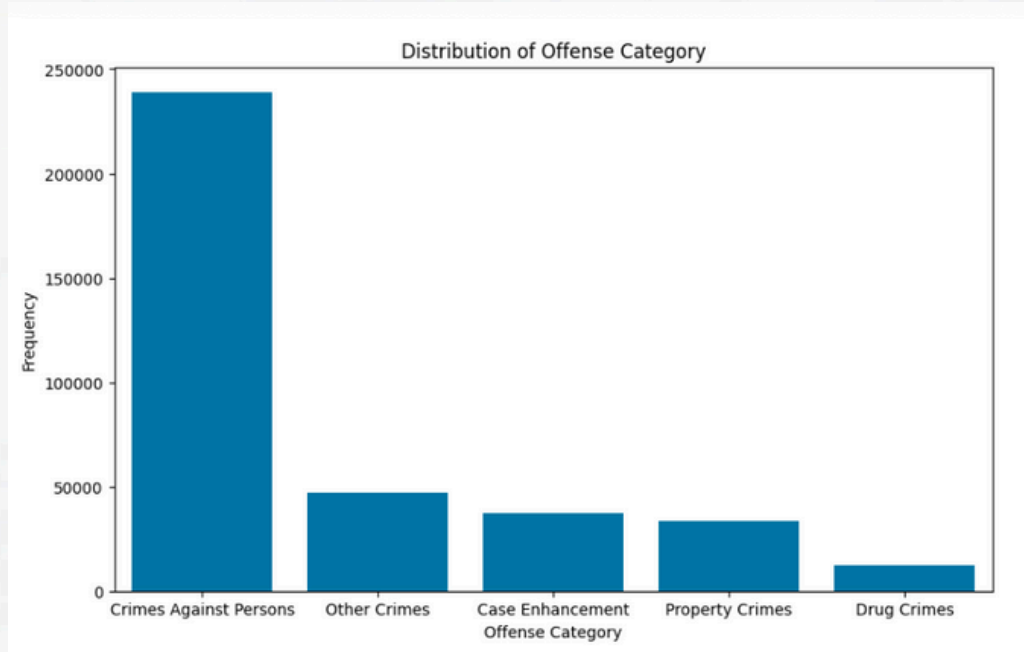
- Filtered out outliers > 2000 months for a clearer view



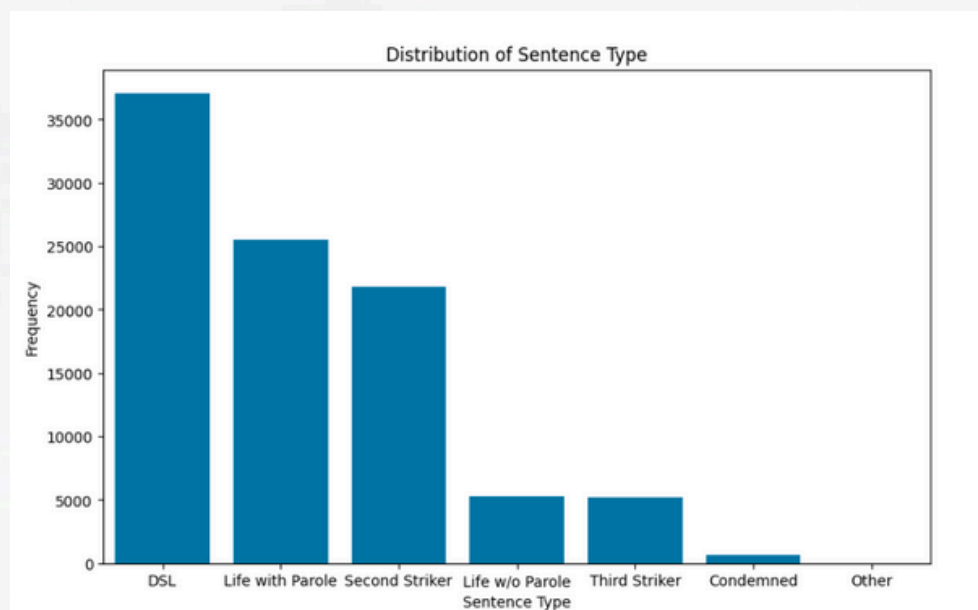
We came to the conclusion that Aggregate Sentence in Months was not a useful feature since a lot of the values turned out to be encodings for different crimes.

DATA VISUALIZATION

- Plot bar plots for
 - Offense Category



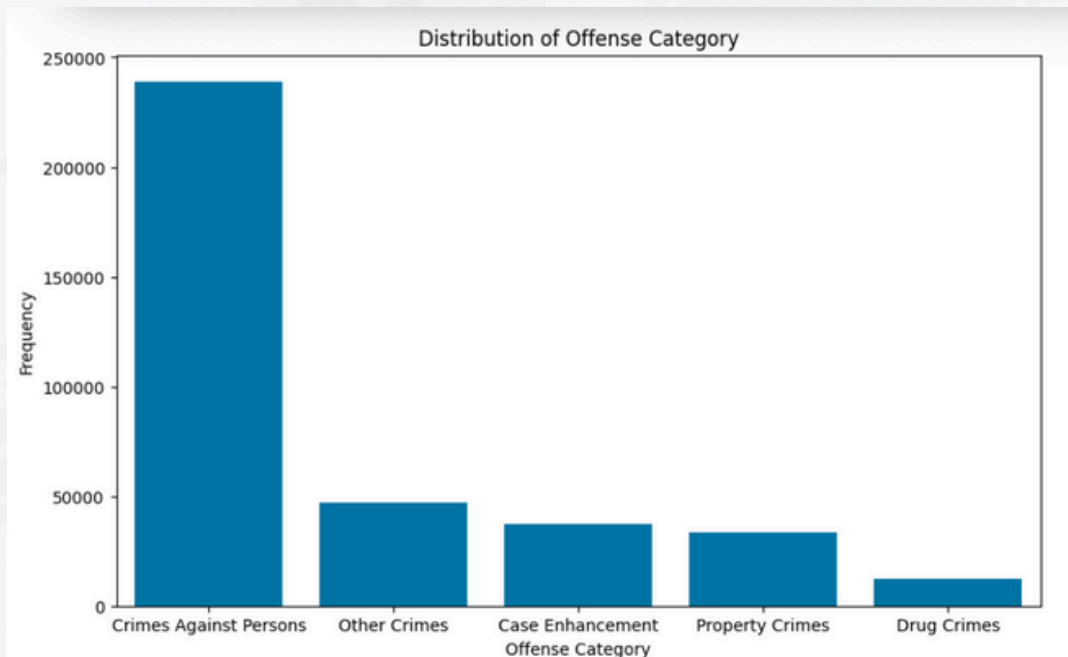
- Sentence Type



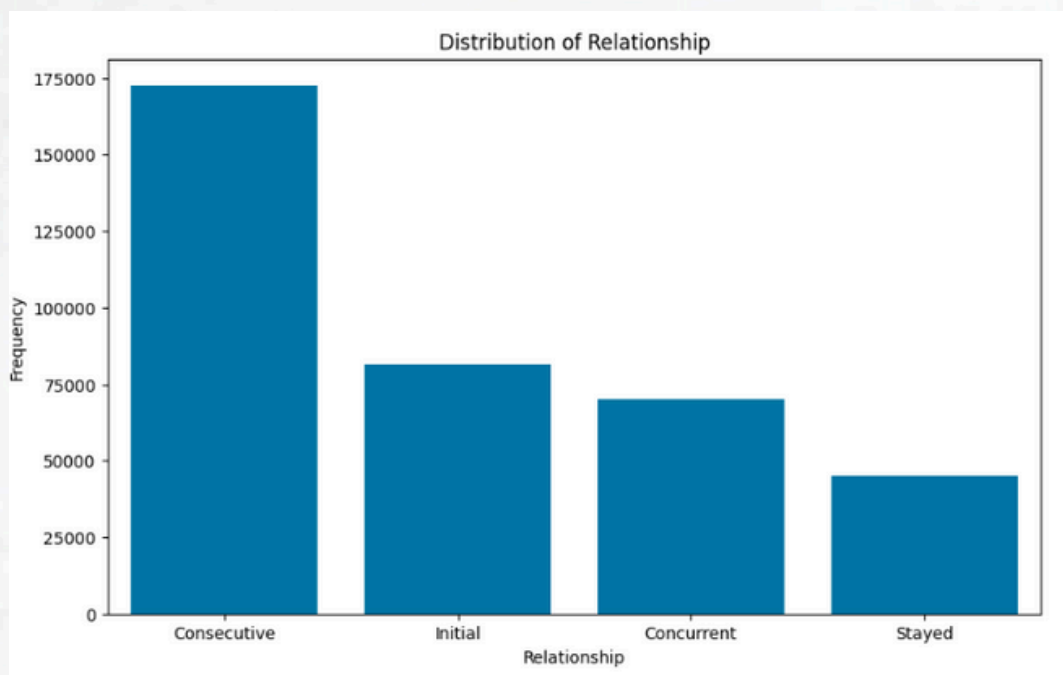
DATA VISUALIZATION

3.1.2 Current Commitments

- Plot bar plots for
 - Offense Category



- Relationship (e.g., principal, accomplice, etc.)



DATA CLEANING

7. Data Cleaning

7.1 Dropping Columns

- Removed unnecessary columns from each dataset:
 - Demographics: e.g.,
 - **Controlling Case Number** (Unless you are tracking a specific legal case, it is just an administrative tracking ID that is not meaningful for analysis)
 - **Current Location** (Represents current prison facility, is not relevant to sentencing trends or offense patterns)
 - Current & Prior Commitments: removed
 - **Case Number**
 - **In-prison**, (Indicates whether the offense occurred while incarcerated. While possibly interesting, it doesn't directly impact sentencing analysis for Three Strikes Law or offense categorization and was excluded to streamline the dataset)
 - Duplicated or less useful sentence data.

7.2 Conversions to Correct Values

- Converted relevant columns from type object datetime:
 - **Offense Begin/End Date, EPRD/MEPD Month and Year, etc.**

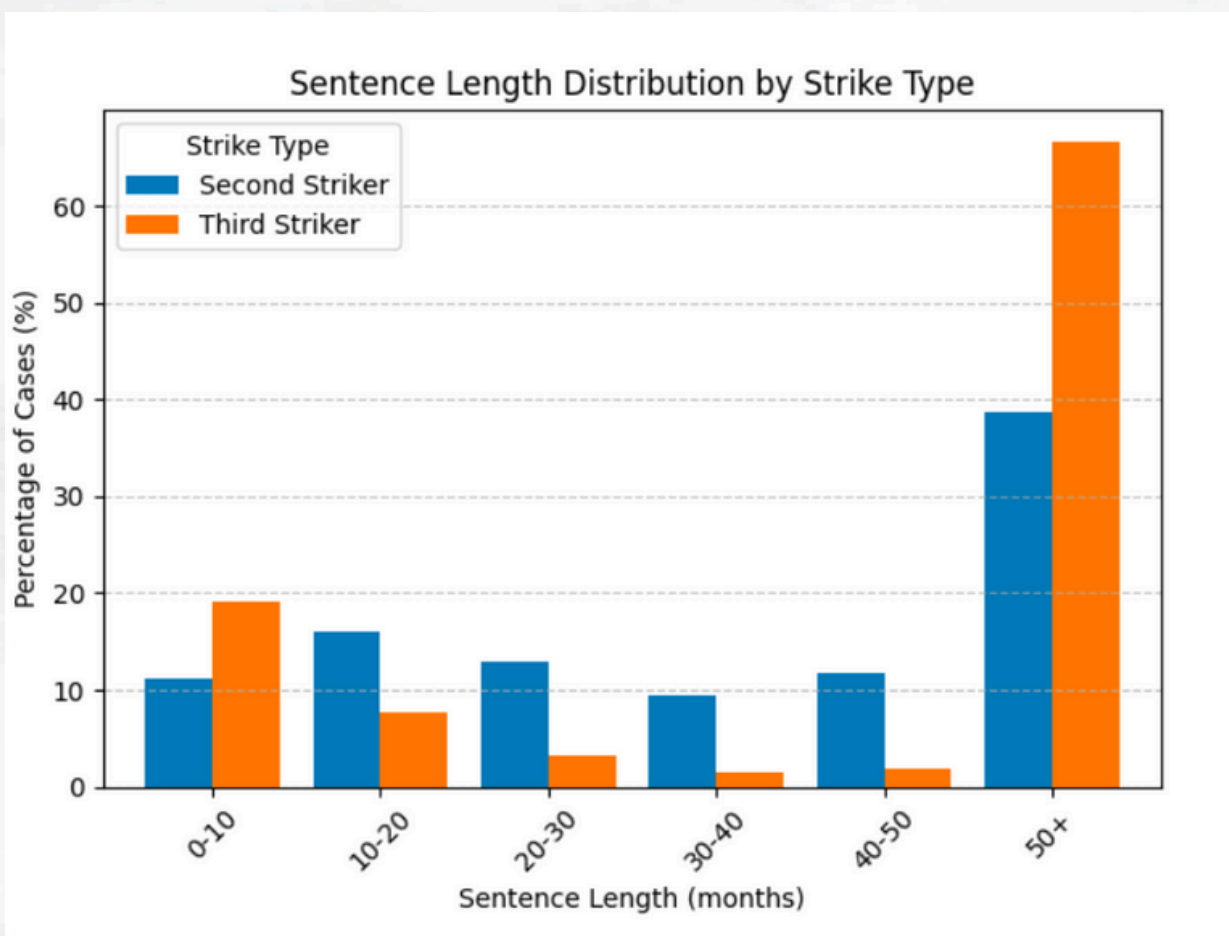
7.3 Sentence Duration Parsing

- Defined **parse_sentence()** to convert string durations (e.g., "2 Years 3 Months") into type int.
- Applied this function to **Offense Time with Enhancement** in both commitment datasets, a column that has proper sentencing times.

ANALYSIS

8. Strike-Based Sentencing

- **Merge:** Join `current_commitments` with `demographics` on `CDCNo` to get strike type.
- **Filter:** Keep only `Second Striker` and `Third Striker`.
- **Bin** sentence lengths (e.g., 0-10, 10-20, ..., 50+ months).
- **Pivot Table:** Compute percentage distribution of sentence length per strike type.
- **Visualization:** Bar chart of sentence length distribution by strike type.



DESIGN EVOLUTION

9. Design Evolution

After the data exploration, we decided to deviate from the original proposal. Originally, we planned to have a time series analysis to show the aggressive measures under Three Strikes, however, after exploring the data, we couldn't find anything that indicated when the individuals were incarcerated. We tried to look for more data sources online for sentencing under Three Strikes outside of our initial proposal, however these attempts were unsuccessful, so we have decided to tweak our proposal to highlight harsh sentencing under Three Strikes and its disproportionate impact on minorities.

Since we were still new to D3, we decided to start with simple visualizations that convey the data well. We started with pie charts to make sure we can properly load and format the data, create proper scales, arcs, and a legend, and added a tooltip to be fancy. We first went with a categorical color scheme that was all primary colors, however we didn't really like the options for categorical color schemes, so we decided to go with blue/purple for all of the graphs. After the first pie chart worked, we created a second one for sentence type.

The next visualization we implemented was the choropleth map outlined in our proposal. We decided instead of doing a proposed adjustable legend where you can view by race or by sentencing, it would be more impactful to make two side by side to show how there are much higher sentencing rates in neighborhoods with a higher percentage of minorities. We continued with the blue/purple schema but used saturation as a channel where higher saturation indicated a higher volume.

For visualization three, which is for sentence length by strike, it was the most complicated one since there was a lot of data manipulation and aggregation that needed to be done. However, we found that we can easily manipulate and format the data in Python, then export it as a CSV and do the rest in Javascript, which is what we did.

IMPLEMENTATION

9.1 PROJECT INTENT

The project aims to provide insight into the effect of California's "Three Strikes and You're Out" law. The interactive visualizations fulfill several purposes:

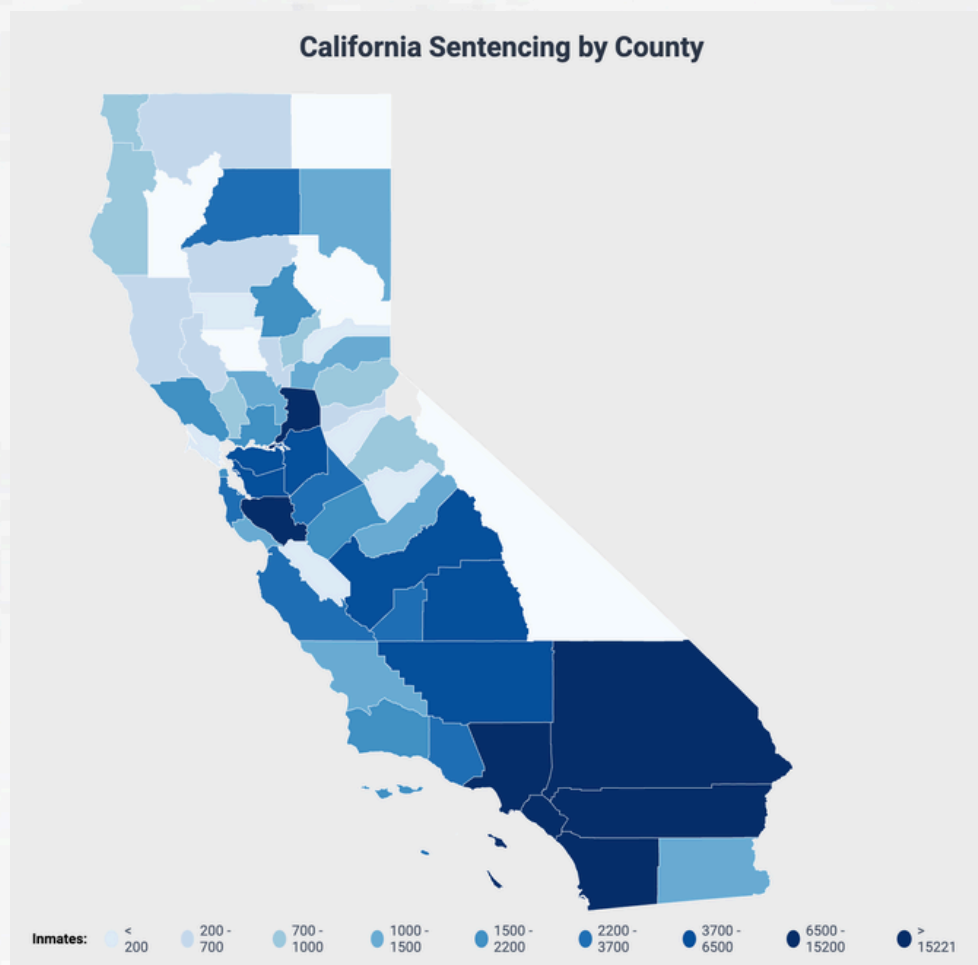
- **Illustrate Geographic Inequities:** Two maps present data at the county level—a map indicating the total number of prisoners and a map indicating minority population percentages. The comparison is intended to encourage discussion about the relationship between demographic factors and sentencing practice.
- **Display Inmate and Sentencing Profiles:** Additional graphs provide breakdowns by inmate ethnicity and sentence type, highlighting judicial outcome disparities.
- **Display Flow and Proportions:** A Sankey diagram is used to illustrate the flow from charges to sentencing results, making it easy to see the distribution of cases across various results.

IMPLEMENTATION

9.2 Key Visualization

9.2.1 California Sentencing by County Map

- **Design Purpose:** This choropleth map uses county boundaries to display inmate counts with a blue color scale.
- **Interaction:** Hovering over a county highlights it and displays a tooltip with the county name and number of inmates.

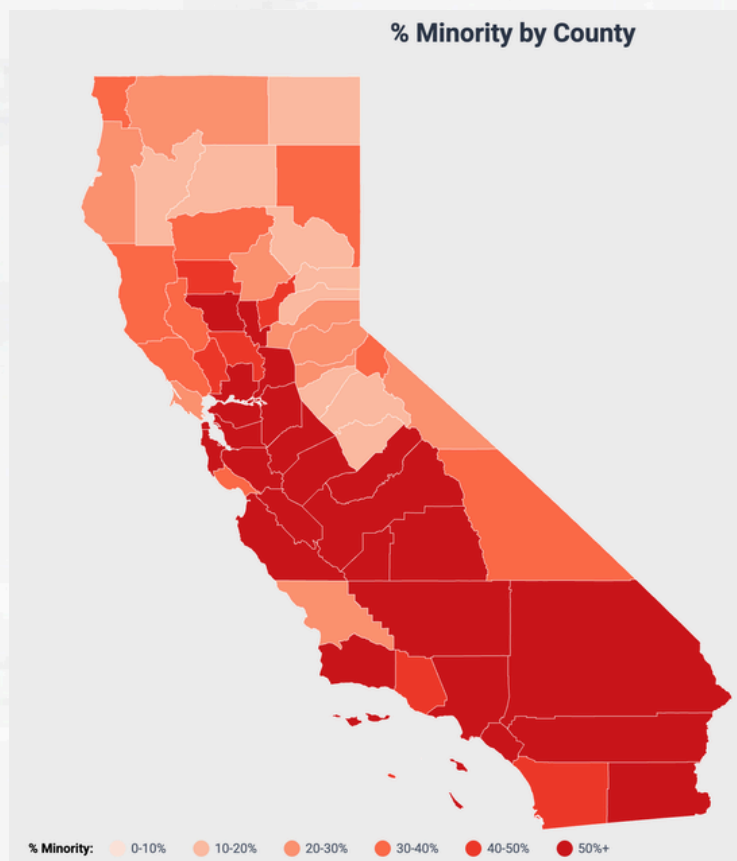


IMPLEMENTATION

9.2 Key Visualization

9.2.2 % Minority by County

- **Design Purpose:** This choropleth map displays all of the California counties color-coded based on the percentage of the population that is in minority groups (combined Hispanic, Black, Asian, Native American, etc.). A light red to dark red color scale is used, with light red for lower percentage and dark red for higher percentage. This layout allows users to quickly identify which counties have higher or lower minority populations, prompting study of potential demographic trends or disparities.
- **Interaction:** Hovering over a county highlights it and displays a tooltip with the county name and breakdown of the minority group.

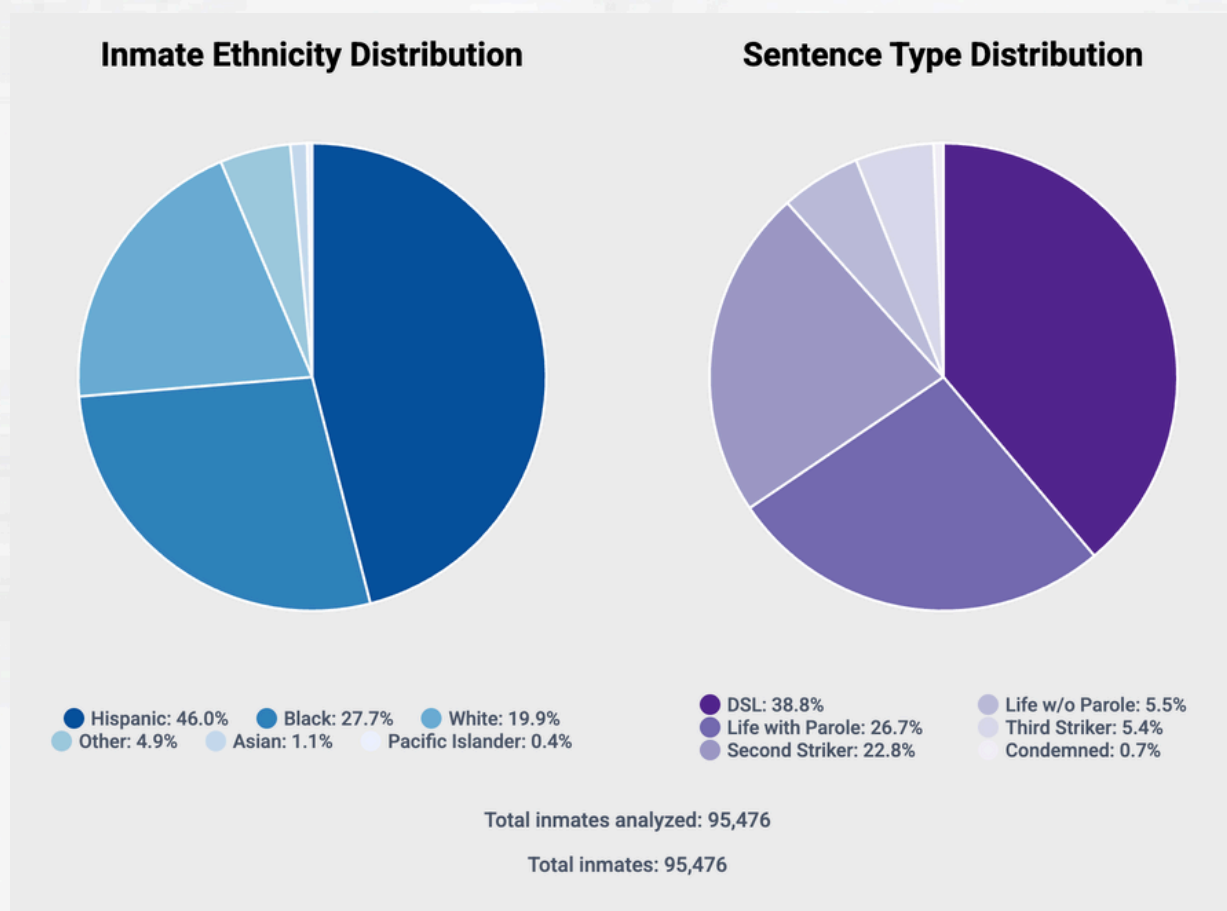


IMPLEMENTATION

9.2 Key Visualization

9.2.3 Inmate Ethnicity Distribution & Sentence Type Distribution

- **Design Purpose:** This pie chart displays the prisoners' ethnicity (i.e., Hispanic, Black, White, Asian, etc.) and sentence type.
- **Interaction:** Users can view precise percentages and figures by hovering over pie slices, which assists in comprehending the demographic spread and their crime.



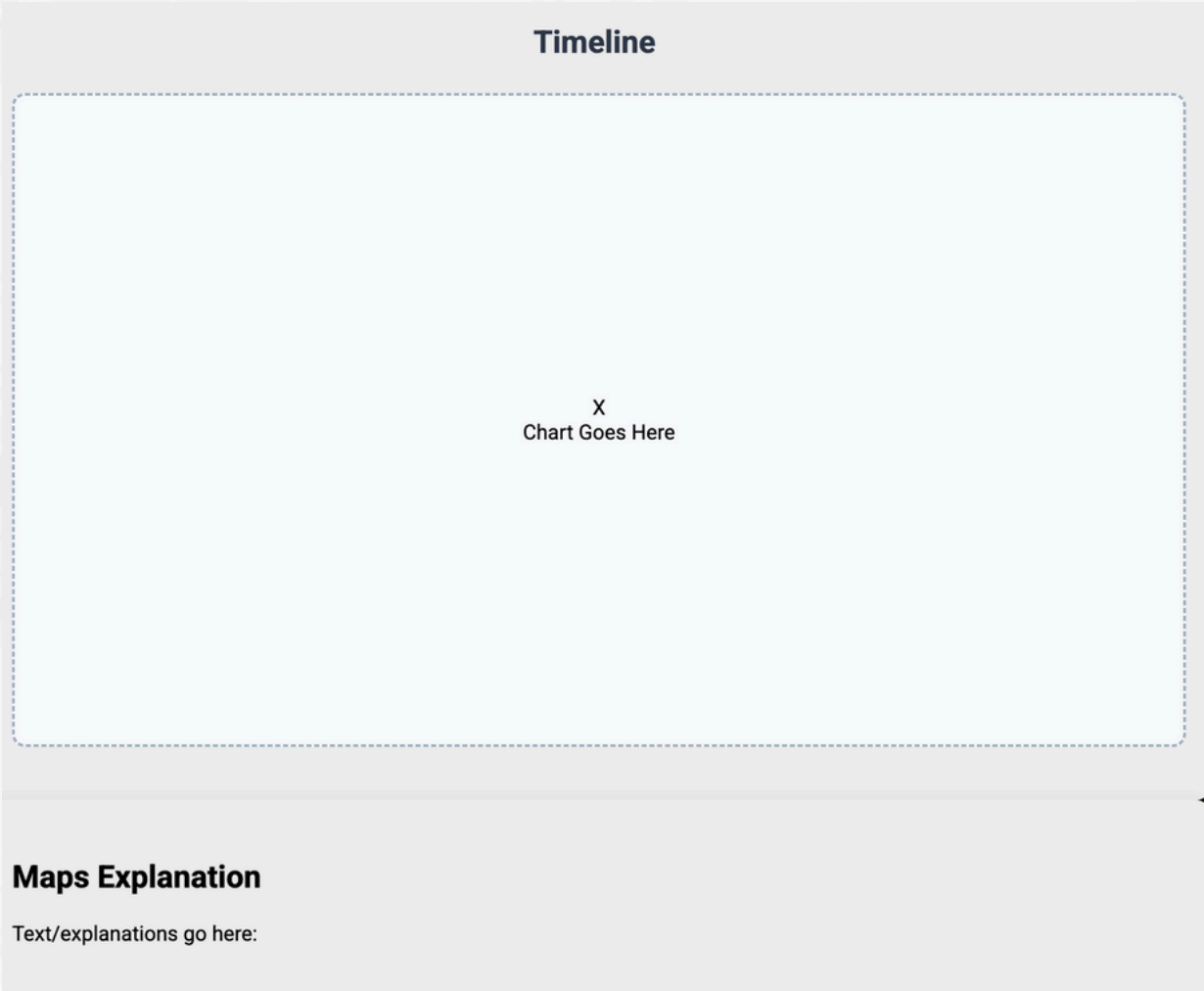
IMPLEMENTATION

9.2 Key Visualization

9.2.6 Timeline Visualization

Design Purpose: The timeline is supposed to show the evolution over time of sentencing and inmate statistics.

Interaction: While represented as a placeholder ("Chart Goes Here") in the PDF, intended functionality involves zooming and filtering by time ranges.



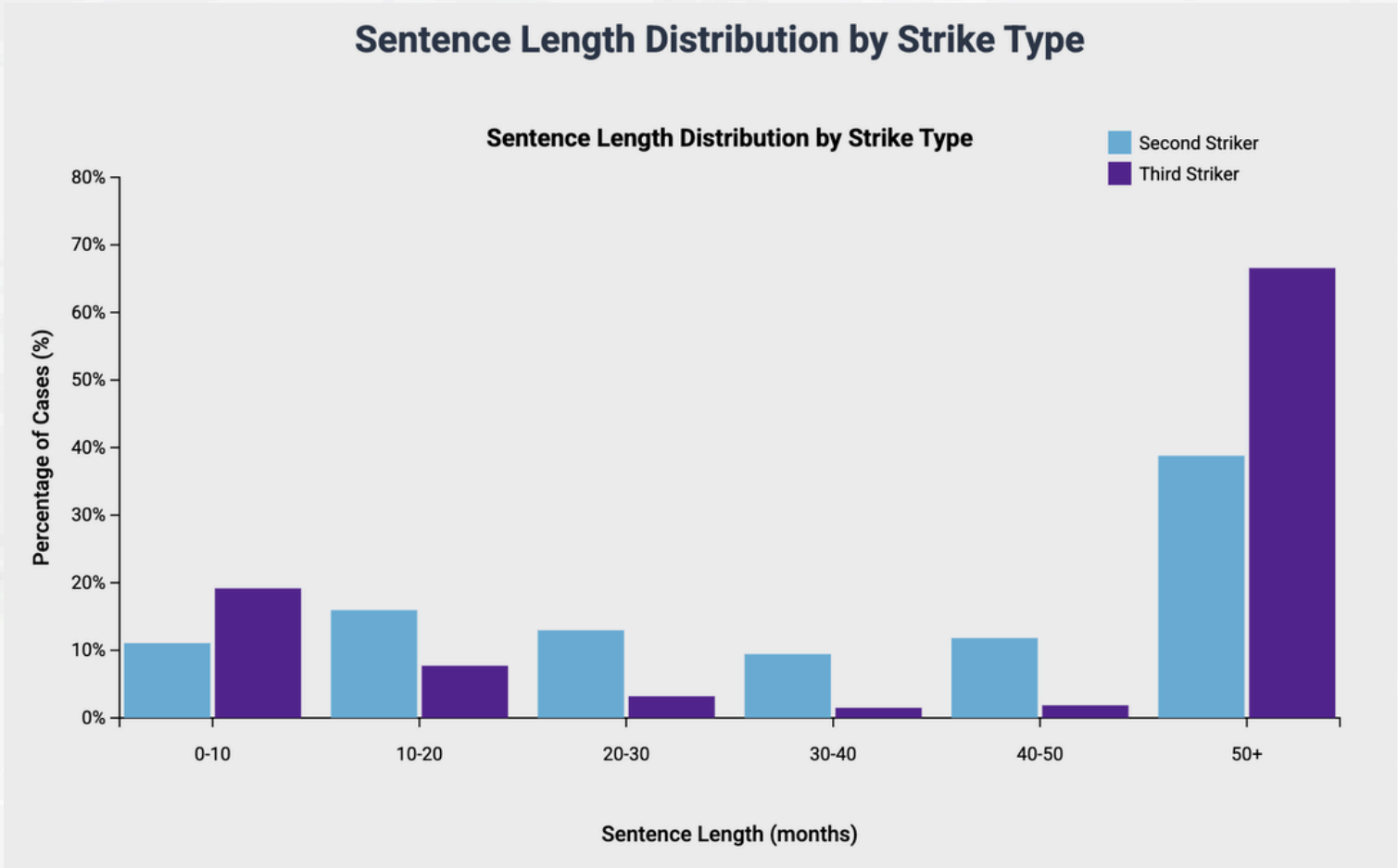
IMPLEMENTATION

9.2 Key Visualization

9.2.4 Sentence Length Distribution by Strike Type

- **Design Purpose:** This graph plots sentence length ranges by strike type, so that users can compare sentencing severity across legal categories.

Interaction: Mouse over the bars will show the label, sentence length, and percentage of cases.



IMPLEMENTATION

9.2 Key Visualization

9.2.5 Sentencing Flow Anlaysis

Design Purpose: The Sankey diagram illustrates the way cases flow through from the original charge to sentencing results, with an emphasis on relative sizes of each pathway.

Interaction: Features like hover and clickable pathways are designed to give a more profound understanding of the sentencing procedure.

Sentencing Flow Analysis

X
Sankey Diagram Goes Here

EVALUATION

Maps Explanation

Text/explanations go here:

EVALUATION

10. Evaluation

10.1 What We Learned About the Data:

We first noticed that our dataset has couple of issues and needed lot of cleaning. For instance, some of our entries were too high, missing, duplicated. There were also numerous columns that was not useful to our analysis so we drop the columns. We ended up finding another dataset that will help determine the minority. We also fix the formatting so the data will reflect to the California sentencing.

EVALUATION

10. Evaluation

10.2 How We Answered Our Research Questions:

We were able to answer our research questions by sorting through and visualizing our data. We began by merging different bits of data (demographics, current and past commitments) so that we could see the entire picture. Then we created a variety of charts and maps—bar charts, pie charts, and choropleth maps to visualize:

- **Patterns of Sentencing:** How long sentences were assigned across different types of strikes.
- **Bias and Disparities:** What population groups were disproportionately impacted?
- **Geographical Trends:** An examination of the variation of sentencing statistics across various counties.

By producing those, it was easier to see the impact of the Three Strikes Law, and to identify trends (e.g., disproportionately severe punishment for minor crimes), and to answer questions regarding who were the most affected individuals and geographical locations.

EVALUATION

10. Evaluation

10.3 Future Improvement:

Our visualization successfully show the trends in data. However, our data is still missing important details like (time information) which was meant to be our initial idea like the timeline. It also didn't include the information on percentage of minority population in each county and to see if there was any relationship between the minority population and sentencing trends, so we ended up using another data to provide the demographic breakdown.

Moving forward, we hope to:

- Make charts more interactive (better tooltips, filters, zooming)
- Gather more complete data (especially related to time)
- Use more models to predict the future sentencing trend